Marcelo Marreiros

Software Engineer

Software engineer with hands-on experience in build systems and platform engineering, currently contributing to the migration of a large-scale retail POS system to a Bazel-based modular architecture used across Sonae's nationwide network. Focused on build reproducibility, developer tooling, and scalable monorepo environments, with strong foundations in C++ and distributed systems.

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WORK EXPERIENCE

Software Engineer
Tlantic

07/2024 - Present — *Porto, Portugal*

www.tlantic.com

Achievements/Tasks

- Contributed to the migration of a large-scale retail POS system from a monolithic architecture to a Bazel-based modular monorepo (Unifo Modularization).
- Developed and maintained Bazel build configurations, improving build reproducibility and consistency across services.
- Worked with protobuf contracts and gRPC integrations to support communication between modular components.
- Built internal tooling for static analysis and automated code formatting, improving developer productivity and code quality.
- Collaborated with cross-functional teams to resolve build and integration issues, improving reliability of development workflows.

Software Development Intern
Tlantic

09/2020 - 07/2021 — *Porto, Portugal*

www.tlantic.com

Achievements/Tasks

- Co-developed a React Native POS app , integrating existing Tlantic services to improve retail operations.
- Participated in client meetings, contributing to design mockups and requirements discussions.
- Collaborated within a cross-functional team to align technical solutions with client expectations.

EDUCATION

Master's in Biomedical Engineering
University of Minho

2016 - 2023

Specialization

- Medical Informatics

SKILLS

C++ · Bazel · gRPC · Protobuf · Python · Javascript · SQL

PERSONAL PROJECTS

FMdeploy - Simplifying Machine Learning Deployment for Academia via Web Interfaces (12/2023)

- Dissertation.
- Developed FMdeploy, a system facilitating the deployment of machine learning models, leveraging the technologies FastAPI and React. This platform aimed to simplify the complex model deployment process through user-friendly interfaces.

Classification of Polycystic Ovary Syndrome Based on Correlation Weight Using Machine Learning (04/2022)

- Book Chapter Published.
- DOI: "10.4018/978-1-7998-9172-7".
- Employed Data Mining techniques (Random Forest) using RapidMiner to develop a model for distinguishing between individuals with PCOS and healthy subjects.

ACHIEVEMENTS

3D Printing Community Impact (12/2023 - 07/2024)

Designed and published 200+ 3D models with over 100,000 downloads across major platforms.

POS System Modularization (08/2024 - Present)

Contributed to the migration of a large-scale retail POS system to a modular Bazel-based architecture.

LANGUAGES

Portuguese
Native

English
Full Professional Proficiency

INTERESTS

3D Printing · DIY Electronics · Software Development · Motorcycles